

STT-1100 - Cheat Sheet

Laval University

```
knitr::opts_chunk$set(echo = TRUE, eval = FALSE, message = FALSE, warning = FALSE)
```

WORK ENVIRONMENT

RStudio

- Script: Ctrl + Shift + N
- Render Quarto: Ctrl + Shift + K

Quarto

```
---  
title: "Title"  
format: html  
---
```

Chunk R:

```
::: {.cell}  
```${.r .cell-code}  
summary(df)
```:
```

Core packages

```
library(tidyverse)
library(readr)
library(UlavalSSD)
library(palmerpenguins)
```

DATA WRANGLING - dplyr

Core verbs

```
select(df, col1)
filter(df, col1 > 10)
```

Transformation

```
mutate(df, z = x/y)
arrange(df, desc(x))
```

Group and summarize

```
group_by(df, cat)
summarise(df, avg = mean(x, na.rm = TRUE))
```

VISUALIZATION - ggplot2

Numerical

```
ggplot(df, aes(x, y)) + geom_point()
ggplot(df, aes(x)) + geom_histogram()
```

Categorical

```
ggplot(df, aes(x, fill = cat)) + geom_bar()
ggplot(df, aes(cat, num)) + geom_boxplot()
```

Use `facet_wrap()` to compare multiple groups.

DESCRIPTIVE ANALYSIS

Numerical summaries

```
mean(x, na.rm = TRUE)
sd(x, na.rm = TRUE)
summary(df)
```

Frequencies

```
table(df$cat)
prop.table(table(df$cat))
```

skimr

```
library(skimr)
skim(df)
```

GOOD PRACTICE TIPS

- `glimpse(df)`: structure preview
- `head(df)` / `tail(df)`
- Comments # to clarify
- Frequent commits in GitHub
- Avoid ambiguous variable names